⊆
H ≤ G subgroup
N ◁ G normal subgroup
a ≠ b
A ∩ B
A ∪ B
∪ A_n
∩ A_n
∏
∤

∅
∈
⊆
χ
τ
ε
η
$\alpha\beta\gamma\delta\varepsilon\zeta\eta\theta\iota\kappa\lambda\mu\varphi\phi$

G/H quotient group
G|S quotient set

==========
proof

QED
==========

Open questions

Q1: self-normal
What can we say about H if it is self-normal?
i.e.
H = N(H)

It is always

aH = bH => H^a = H^b

but NOT <=, EXCEPT when self-normal.

Self-normal means 1-1 relationship

aH <=> H^a

Q2: normal series
Consider normal series

H ◁ N(H)
N(H) ◁ N^2(H)
N^2(H) ◁ N^3(H)
....

Then eventually, either

N^n(H) = G or
N^n(H) self-normal

When?

Basics

NOTE: the main idea is that for any a, b ∈ N, **linear combinations** of a and b cannot be smaller than (a,b),
or more precisely, always a multiple of (a,b)

(a,b) | ua + bv for any u,v

Division Theorem
Let a, b ∈ N. There exists unique non-negative integers q, r such that

a = qb + r r < b

Theorem (GCD)
Let a, b ∈ N.

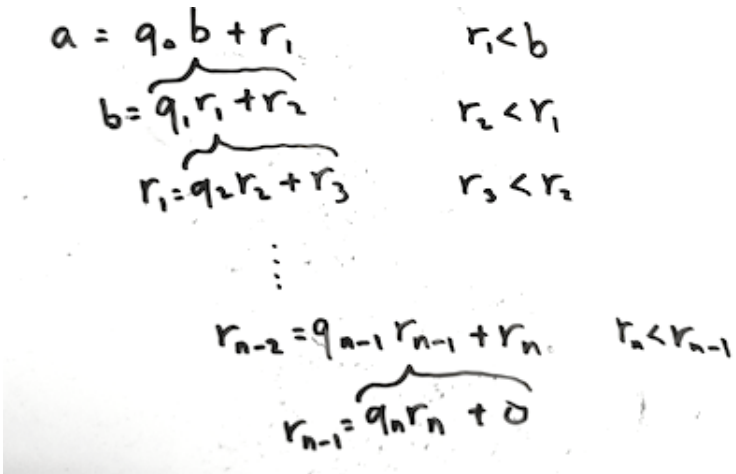
a) **fundamental property**: Any linear combination of a,b is a factor of (a,b)

(a,b) | ua + bv any u,v

Furthermore, there exists u,v such that

(a,b) = ua + vb

b) **computation**: (a,b) can be computed systematically by the Euclidean algorithm.
Compute the division sequence



Then rn = (a,b)

==========

proof
a)
Consider the set of all possible x = ua + vb.
Let d be the **smallest**.

Claim: d | x
Suppose on the contrary

d ∤ x

=> x = qd + r for some r < d
=> r = x - qd
= (ua + vb) - q(u'a + v'b)
= (u-qu')a + (v-qv')b
=> contradicts that d is smallest

QED claim

Then

d | a = 1a + 0b => d <= (a,b)
d | b = 0a + 1b

d = ua + bv => (a,b) | d
=> (a,b) <= d

b)

(a,b) | a, b => (a,b) | r1 since a = qb + r1
=> (a,b) | r2 since b = q1r1 + r2
=> (a,b) | r3 ...
...
=> (a,b) | rn

rn | rn-1 => rn | rn-2 since rn-2 = qn-1rn-1 + rn
=> rn | rn-3
=> ...
=> rn | b since b = q1r1 + r2
=> rn | a since a = q0b + r1
=> rn | (a,b)

Hence (a,b) = rn

QED
==========

Theorem (Euler's totient function)

$\phi(m) = m \prod_{p \mid m} (1-1/p)$ simply since a) $\phi(p^k) = p^k - p^{(k-1)} = p^k (1-1/p)$

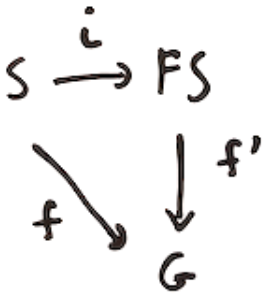
b) $\phi(nm) = \phi(n)\phi(m)$ for $(n,m) = 1$

Free Group

Universal property

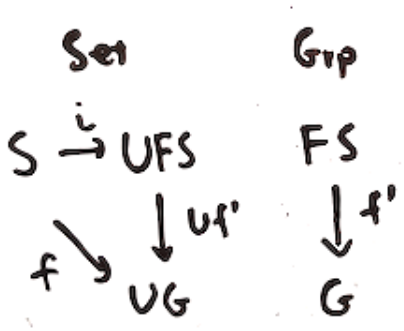
Set		**Group**
function		homomorphism
f: S → G	<=>	f: FS → G
	f = f'	

with **injection** i: S → FS.



In category theoretic terms:

injection as **universal map**	i: S → UFS
Free functor	F: S → FS for any set S
Forgetful functor	U: G → UG for any group G
Adjoint	{ f: S → UG } <=> { f': FS → G }
	Set Grp



==========

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== Groups ==

Lagrange's Theorem
If H is a subgroup of a finite group G, then |H| divides |G| and

[G:H] = |G|/|H|

proof: by disjointness of the cosets

Euler's theorem (+**Fermat's little theorem)**
If (a,m) = 1, then

a^ϕ(m) = 1 mod m

In particular, if m is prime, then

a^p-1 = 1 mod p (Fermat's little theorem)

==========

proof

(a, m) = 1
=> a ∈ Zm*
=> |(a)| | |Zm*| = ϕ(m)
i.e. order of cyclic subgroup (a) divides order of group
=> |a|^k = 1 mod m

QED

==========

Theorem: canonical representation
Let G be a group, H ◁ G and * an action on G|H

*: G x G|H → G|H
g * xH = gxH

Then

H contains **no normal** subgroup of G

G	≃	**Im** *
g	<=>	g*: G|H → G|H ∈ S(G|H)
	gH = g*(H)	

H is **normal**
i.e. G/H is a quotient group

G/H	≃	**Im ***
gH	<=>	g*: G/H → G/H ∈ S(G/H)
	gH = g*(H)	

Cayley's theorem (H = G)
Each g ∈ G is characterized by its action on the other elements

G	≃	**Im *** ⊆ **S(G)**
g	<=>	g*: G → G ∈ S(G)
	g = g*(1)	

Hence for any |G| = n and G = { g0 = 1, g1, g2, ... }

G	≃	**H = { φi: Sn → Sn }** subgroup of Sn where for each i ∈ G, there is **exactly one** φi(1) = i
gi	<=>	φi: Sn → Sn
	gi = φi	

==========

Proof
or any g* ∈ ker *

H = g*(H) = gH

=> g ∈ H
i.e. ker * ⊆ H
=> ker * is trivial since ker * is normal

=> Im * ≃ G/ker * = G

QED
==========

Product
https://en.wikipedia.org/wiki/Product_of_group_subsets

Theorem: products
Let H, K be subgroups of G

a) existence:
The canonical condition is

HK = KH <=> **HK** is a subgroup

Some common cases are

normality

H or K normal => **HK** subgroup since HK = KH by normality

H ∩ **K** normal

H and K normal => **HK** normal since (HK)^g = H^gK^g = HK

In particular, this is relevant for **quotient groups**.
See the quotient group section for details.
e.g.

H = HK subgoup of G <=> **H'/K** subgroup of G/K
H = H'K

H = HK normal in G <=> H'/K normal in G/K

Special case is when **K = (a)** i.e. a single element

H is **normal** <=> **H(a)** is a subgroup for all a ∈ G
i.e. N(H) = G i.e. H can be **extended** by a single element

H is **self-normal** <=> **H(a)** is NOT a subgroup for any a ∈ G
i.e. N(H) = H i.e. H cannot be **extended** by any single element

b) **order**: If HK is a subgroup

$$|HK| = |H||K| / |H \cap K|$$

This is because for any $h,h' \in H$, $k, k' \in K$

$hk = h'k'$ <=> $h' = hx$ $x \in H \cap K$
$k' = x^{-1} k$

Semi direct product
https://en.wikipedia.org/wiki/Semidirect_product

Semi direct product
Let G be a group, and N, H be subgroups of G with N normal.
Then the product subgroup NH exists and

$(nh)(n'h') = n(hn'h^{-1})hh'$

$= (n\, n'^h) \cdot (hh')$
$\in N \quad \in H$

N x H can be made into a group

inner semi direct product N x H
element $(n, h) \in N \times H$

operation $(n, h) \circ (n', h') := (n\, n'^h, hh')$

note: If $H \subseteq C(N)$ i.e. if H commutes so $n^h = n$
then this is the usual **direct product**

$(n, h)(n', h') = (nn', hh')$

There is a group homomorphism

$\varphi: N \times H \to NH$
$\varphi(n, h) = nh$ note: NOT always an isomorphism, since each nh need not be unique.

NH is called a **semi product N $\rtimes$ H** if this is an isomorphism.

Theorem: semi direct product definition
Let G be a group, and N, H be subgroups of G with N normal.

Equivalent definitions of semi direct product:

a) **decomposition N x H**: NH can be **decomposed** as N x H
i.e. inner semi direct product N x H is isomorphic to NH with map

$\varphi: N \times H \to NH$
$\varphi(n, h) = nh$

b) **complement**: $N \cap H = \{ 1 \}$

c) **unique factorization**: every $x \in NH$ can be uniquely decomposed as x =

nh

d) **quotient**: the quotient map π: G → G/N is an isomorphism on H
i.e. π|H: H → H/N is an isomorphism

e) **homomorphism**: there is a homomorphism φ: G → H such that
φ|H=id and ker φ = N.

==========

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== Coset ==

Coset
https://en.wikipedia.org/wiki/Coset
Let G be a group and H a subgroup.
Then

left coset aH := { ah: h ∈ H }

right coset Ha := { ha: h ∈ H }

conjugate H^a := { aha^-1: h ∈ H }

Theorem: Basic properties
Let H be a subgroup of G

uniqueness

Each ah ∈ aH is unique

ah = ah' <=> h = h'

Hence it has the same order as |H|

|aH| = |H|

abelian

aH = Ha <=> H^a = H

i.e. **invariance** under conjugation is equivalent to weak commutative

equivalence

aH = bH <=> ba^-1 ∈ H

Hence the cosets for an **equivalence relation**

partition

The left cosets partition G by the equivalence relation, each of the same order

aH ∪ bH = aH or ∅ i.e. disjoint

|aH| = |H| order

G = ⊎ aH disjoint union

[G:H] = |G| / |H|

Any b ∈ aH can be used as a **representative**

aH = bH <=> b ∈ aH

identity

b(aH) = aH <=> b ∈ H^a **identity**

b(aH) = H <=> b ∈ Ha^-1 **inverse**

Hence a 1-1

G * aH ≃ G|H^a
b(aH) <=> b(H^a)

If N is a **normal** subgroup, then this is just

b(aN) = aN <=> b ∈ N since N^a = N

b(aN) = N <=> b ∈ a^-1N since gN = Ng

power

(a^k)H = (a^l)H <=> a^l-k ∈ H

Hence, if a^n ∈ H is the smallest such n, then

aH, a^2H, ..., a^n-1 H, a^nH = H

are **unique**.

representation

Each g ∈ G can be defined by how it **maps** each coset aH
i.e. by the permutation g * : G|H → G|H
For simplicity, write

g(a) = b <=> g(aH) = bH

Then

g(a) = b <=> g ∈ bHa^-1

g(1) = b <=> g ∈ bH **identity**

g(a) = a <=> g ∈ H^a **fixed point**

Partition: The **images G * a** ⊆ **G|H** also **partition** the mappings

g: G * a → G * a
g(x) = g(g'^-1(a)) where x = G'(a)
= (g g'^-1)(a) ∈ G * a

Conjugate & coset

Conjugacy and cosets are closely related
By the usual actions

***: G x G|H → G|H** **^: G^H x G → G^H**
g * aH = gaH H^a ^ g = H^(ga)

If H is a subgroup

commutativity

H^a = H <=> aH = Ha

kernel

b ∈ H^a <=> b(aH) = aH

Equivalently

ker *aH = H^a

Hence, H^a is like an **identity element** for aH

H^a · aH = aH ~ 1 · x = x

H^a · H^a = H^a ~ 1 · 1 = 1

However, there may be **multiple** aH associated with the same H^a:
since **H ⊆ N(H)** and

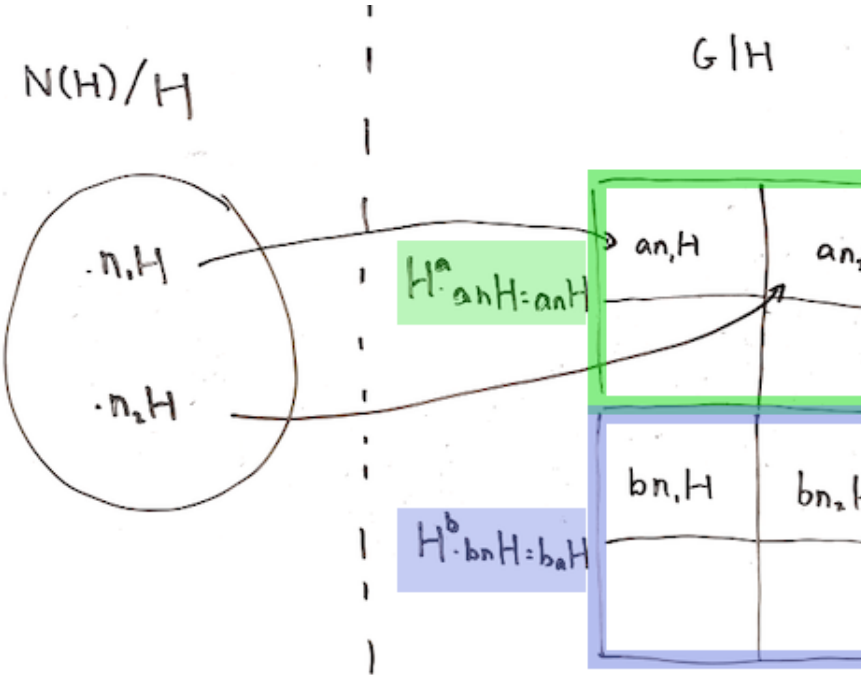
aH = bH <=> b ∈ H

H^a = H^b <=> b ∈ N(H)

This is determined by **N(H) * H** i.e. equivalently **N(H) / H**
This forms **partitions a(N(H) * H)** for each **unique H^a**.

a(N(H) * H) = { anH ∈ G|H | n ∈ N(H)/H }
≃ **N(H) / H**

= { bH ∈ G|H | H^b = H^a } by the equivalence relation
i.e. ker *bH = H^a



Firstly, **H^a** and **aH** are like duals:

ker *aH = H^a

space	**G|H**	-->	**H^G**
element	a ↦ **aH**	--> **ker *aH = H^a**	a ↦ **H^a**
kernel	ker *aH = H^a		ker (H^a)^ = N(H)
equivalence	aH = bH iff b ∈ aH		H^a = H^b iff b ∈ aN(H)
class size	|aH| = |H|		|H^a| = |H|
# classes	|G|H| = |G| / |H|		|H^G| = |G| / |N(H)|

As permutations

element	**π: G → S(G|H)** πg = g*: G|H → G|H πg(aH) = (ga)H	**σ: G → S(H^G)** σg = ^g: H^G → H^G σg(H^a) = H^(ga)
kernel	ker π = Core (H)	ker σ = Core(N(H))

In turn, **H^a** and **aN(H)** are equivalent:

space	**H^G**	≃	**G | N(H)**
element	a ↦ **H^a**	<=> **H^a = H^b**	a ↦ **aN(H)**

iff
b ∈ aN(H)

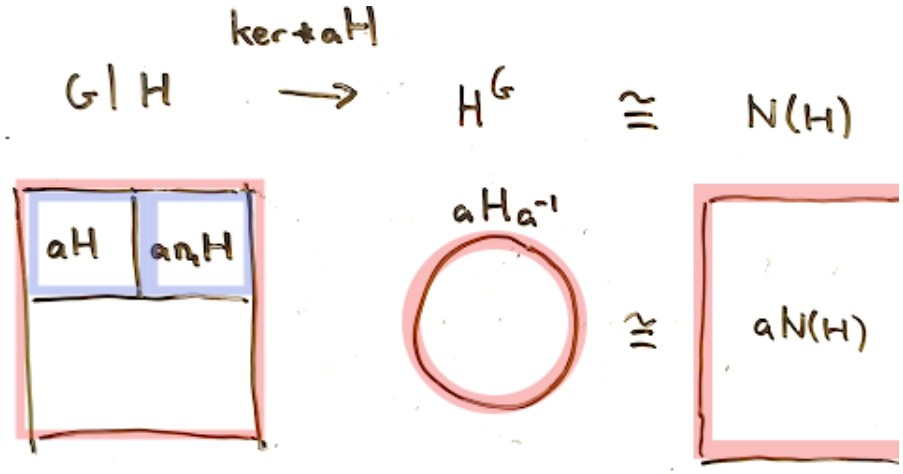
kernel	ker (H^a)^ =	N(H)^a	= ker * aN(H)
class size	|H^a| = |H|		|aN(H)| = |H|
# classes	|H^G| =	|G| / |H|	= |G|N(H)|

as permutations

element	σ: G → S(H^G)	π: N(H) → S(G|N(H))
	σg = ^g: H^G → H^G	πg = g*: G|N(H) → N(H)
	σg(H^a) = H^(ga)	πg(aN(H)) = (ga) * N(H)
kernel	ker σ = Core(N(H))	= ker π

This means if H = N(K) for some K, then there are two equivalent views:

partition	aH
group	K^a

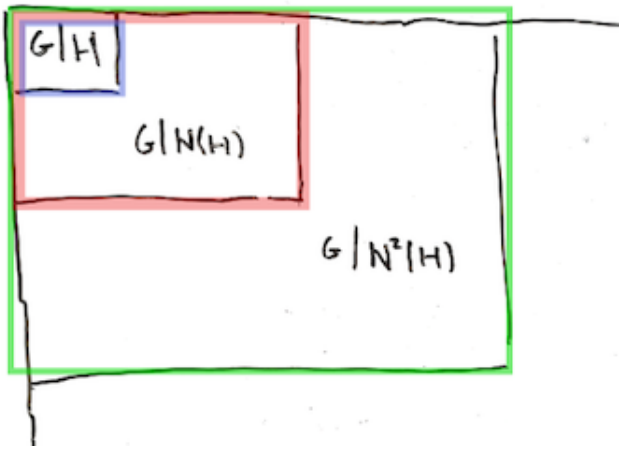


Combining this, we have a **sequence** of relations for **aH, H^a** and **aN(H)**:

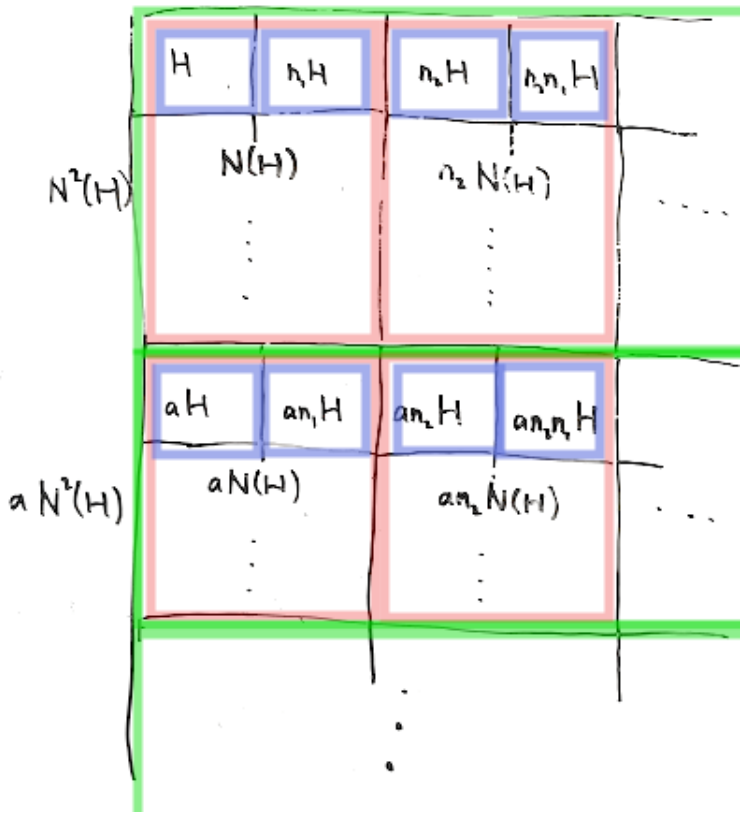
H ◁ N(H) ◁ N^2(H) ◁ N^3(H) ...

G|H => **H^G ≃ G|N(H)** => N(H)^G ≃ G|N^2(H) => N^2(H)^G ≃ G|N^3(H) =>
....

ker *aH = H^a ~ aN(H) ker *aN(H) = N(H)^a ~ aN^2(H)
ker *aN^2(H) = N^2(H)^a ~ aN^3(H)



$$N \triangleleft N(H) \triangleleft N^2(H) \ldots$$



Two extremes are

H normal **H^G = { H }**
i.e. N(H) = G

H self-normalizing **H^G = G|H**
i.e. N(H) = H

Thus, H^G represents how close G|H is to a **quotient group G/H**.
Indeed

G/H is a well-defined quotient group <=> H^G = { H } i.e. H is normal

There may be **multiple aH** corresponding to the same H^a.

Two extremes are

H normal i.e. N(H) = G	**N(H) * H = G|H**	i.e. all cosets
H self-normalizing a ∈ G i.e. N(H) = H	**N(H) * H = { H }**	i.e. singleton for all

Thus, H^G represents how close G|H is to a **quotient group**.
Indeed

G/H is a well-defined quotient group	<=>	H^G = { H } i.e. H is normal

==========

== Homomorphism ==

Group homomorphism:

Homomorphism
Let G, H be groups
A map f: G → H is a group homomorphism if

identity	f(1) = 1
group operation	f(ab) = f(a)f(b)

Immediate consequences

inverse	f(a^-1) = f(a)^-1

Special homomorphisms

endomorphism	f: G → G
monomorphism	f: G → H injective i.e. Ker f = {1}
epimorphism	f: G → H surjective i.e. Im f = H
isomorphism	f: G → H bijective i.e. mono + epi
automorphism	f: G → G bijective i.e. endo + iso

Theorem: Isomorphism
Let G, H be groups, f: G → H a homomorphism

image, kernel	Im f is a **subgroup** of H
	Ker f is a **normal subgroup** of G
fundamental property	**G/Ker f ≃ Im f**
	Hence f can be decomposed as
	f = $\alpha\beta\gamma$
	γ: G → G/Ker f surjection
	β: G/Ker f → Im f isomorphism
	α: Im f → H injection

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== Group action ==

Group action: https://en.wikipedia.org/wiki/Group_action

Group action *
Let G be a group and X a set of elements.
An **action on X by G** is a map *** : G x X → X** where

1 * x = x	identity
g * (h * x) = (gh) * x	group operation

Given x, the **image and kernel of *x** (aka orbit and stabilizer of x) are

image: orbit of x
G * x = **Im** * x
= { g * x ∈ X : g ∈ G }

kernel: stabilizer of x
Gx = **ker** * x
= { g ∈ G : g ∈ G }

Gx is always a **subgroup**

If y = g * x, then

S * y = Sg * x for any subset S ⊆ G

Gy = (Gx)^g
i.e. conjugate g(Gx)g^-1

Values of actions
Each g * x need not be unique.
In particular, there may exist **fixed points x**

g * x = x

for g ≠ 1.

The stabilizer characterizes an equivalence

g * x = h * x <=> g ~ h mod Gx
g = hg for some g ∈ Gx

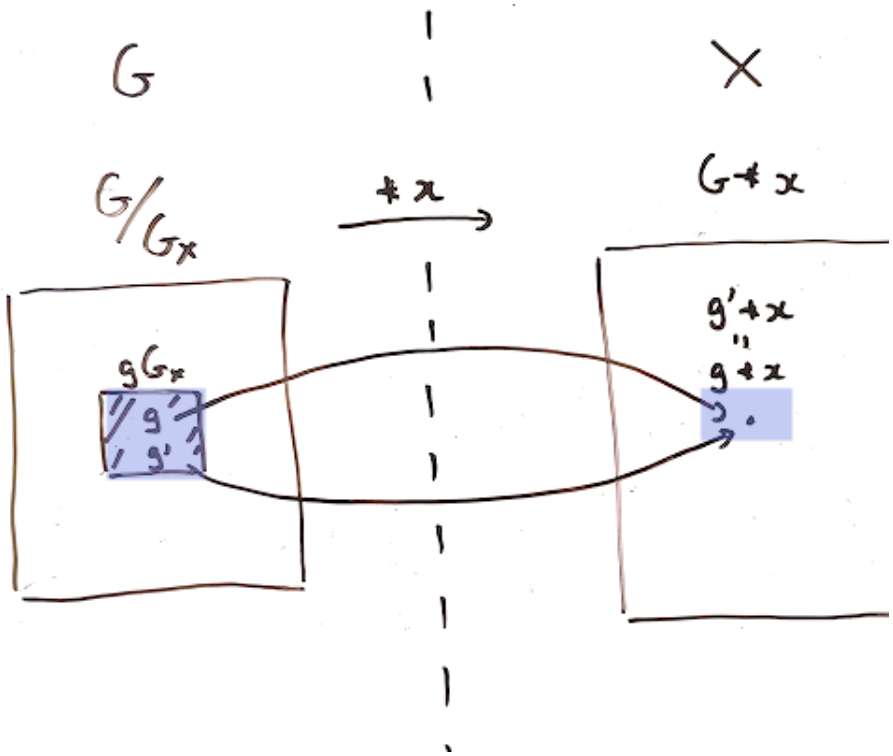
Hence

a) 1-1 correspondence:
Hence, unique values of g * x have a 1-1 correspondence with **quotient set G|Gx**
note: G/S = quotient group G|S = quotient set

	X		G
set	G * x	≃	G|Gx
element	x g * x	<=>	Gx gGx
equivalence	g * x = h * x	<=>	g ~ h mod Gx

Or as a bijection (note: not a group homomorphism)

G|Gx ≃ G * x
G|Gx → G * x
gGx ↦ g * x



b) **order** (orbit stabilizer theorem):
By the 1-1 correspondence, the number of unique values of g * x are

$$|G * x| = |G|Gx| = |G| / |Gx|$$

Interpretation

There are 2 dual ways to view actions:

1) **$* x: G \to X$**
i.e. **actions** on a fixed $x \in X$

Each G * x is

$$G * x \simeq G|Gx$$
$$g * x \iff gGx$$

Furthermore, each G * x is precisely an **equivalence classes** mod G of X

x ~ y mod G $\iff$ y = g * x for some $g \in G$

$\iff$ **$y \in G * x$**

Hence, the images aka orbits **partition** X

partition $G * x = G * y$ or $G * x \cap G * y = \varnothing$

$G * x = G * y \iff y = a * x$ for some $a \in G$

kernel aka stabilizer $G * x = G * y \implies Gy = Gx^a$ where $y = a * x$
i.e. kernel are **conjugates**

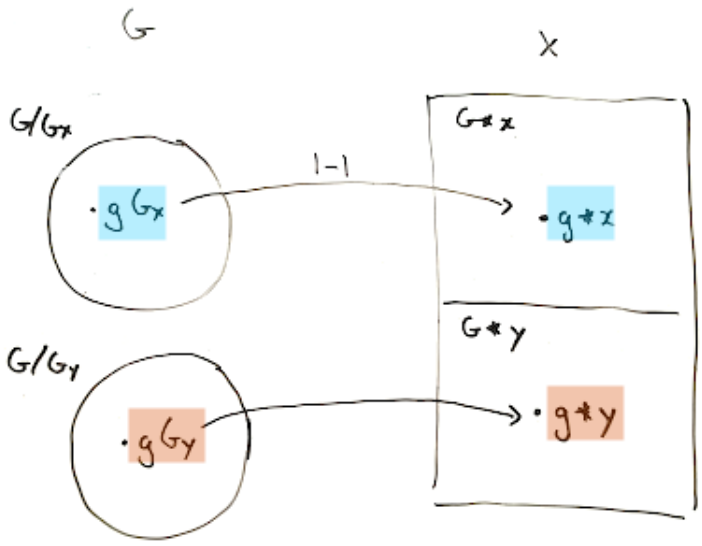
quotient set
$X / G = \{ G * x : x \in X \}$
i.e. set of all classes

Then

$$X = \biguplus_{X/G} G * x$$

class equation

$$|X| = \sum_{X/G} |G * x| = |G| \sum_{X/G} 1 / |Gx|$$



2) **representation** of G as πg = **g*: X → X**
Each g ∈ G can be thought of as characterized by its relationship with each x ∈ X

g <--> gx0, gx1, gx2, ... for all xi ∈ X

i.e. represented as the permutation πg = **g*: X → X**

uniqueness
g * x = g * y <=> **x = y**

Thus, each g ∈ G has a 1-1 correspondence with a **permutation** .

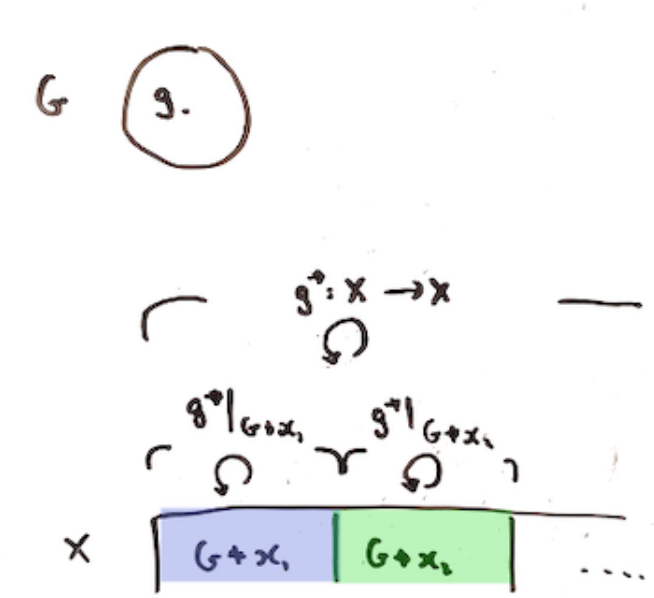
π: G → S(X)
π(g) = g*

g ∈ G => σ ∈ S(X)
g * x = σ(x)

G / ker π ≃ subgroup of S(X)
g(ker π) πg: X → X

Each image G * x **partitions** the permutations.
i.e. each can be restricted to G * x

πg: G * x → G * x
πg (a * x) = (ga) * x ∈ G * x



Action vs group homomorphism
An action can be **induced** by a group **homomorphism**
e.g.
Let f: G → H be a homomorphism.
Define action

*: G x H → H
g * h = f(g)h

Then

***1 = f**
i.e. g*1 = f(g)

ker *h = h (ker f)
Im *h = h (Im f)

Examples

	Action *: G x X → X	**Class**
G * x		
Action on G	*: H x G → G	class
Im ^a ≃	H | ker ^a	
by conjugation h ↦ a^h	h * a = a^h	
= a^H	= H | H ∩ N(a)	
	ker a^ = **H** ∩ **N(a)**	

element **h ↦ a^h** **<=>** **h ↦ h(H ∩ N(a))**

|G|

= ∑ |a^H|

$= \sum |H| / |H \cap N(a)|$

$= \sum 1 \quad + \quad \sum |H| / |H \cap N(a)|$

over $H \subseteq N(a)$ rest

Action on subgroup K $*: H \times K^G \to K^G$ class

$\text{Im } *K^a \simeq H \mid \ker *K^a$

by conjugation h ↦ K^a $h * K^a = (K^a)^h$

$= K^{Ha}$ $= H \mid (H \cap N(K)^a)$

$= K^{(ha)}$

$\ker * K^a = H \cap N(K^a)$

element **h ↦ K^ha** **<=>** **h ↦ h(H ∩ N(K)^a)**

= H ∩ N(K)^a

class

size

$|K^G|$

$= \sum |K^{Ha}|$

$= \sum |H| / |H \cap N(K)^a|$

note:

$= \sum 1 \quad + \quad \sum |H| / |H \cap N(K)^a|$

$K^G \simeq G|N(K)$

over $H \subseteq N(K)^a$ rest

$a \mapsto K^a \qquad a \mapsto aN(K)$

Action on G|K $*: H \times G|K \to G|K$ class

$\text{Im } *aK \simeq H \mid \ker *aK$

$h * aK = haK$

$= H \mid aK$ $= H \mid H \cap K^a$

$\ker *aK = \mathbf{H \cap K^a}$

element **h ↦ h(aK)** **<=>** **h ↦ h(H ∩ K^a)**

class

size

$|G|K|$

$= \sum |H * aK|$

$= \sum |H| / |H \cap K^a|$

$= \sum 1 \quad + \quad \sum |H| / |H \cap K^a|$

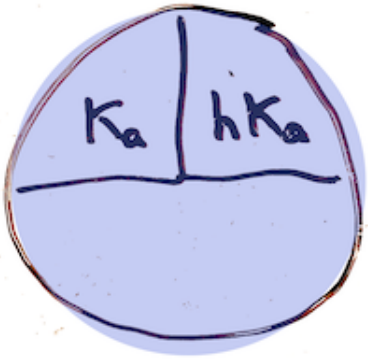
over $H \subseteq K^a$ rest

Image $H \mid aK \simeq H / H \cap K^a$

H

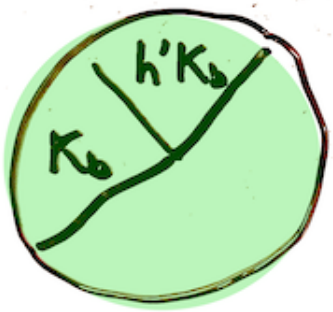
$H \mid K_a$

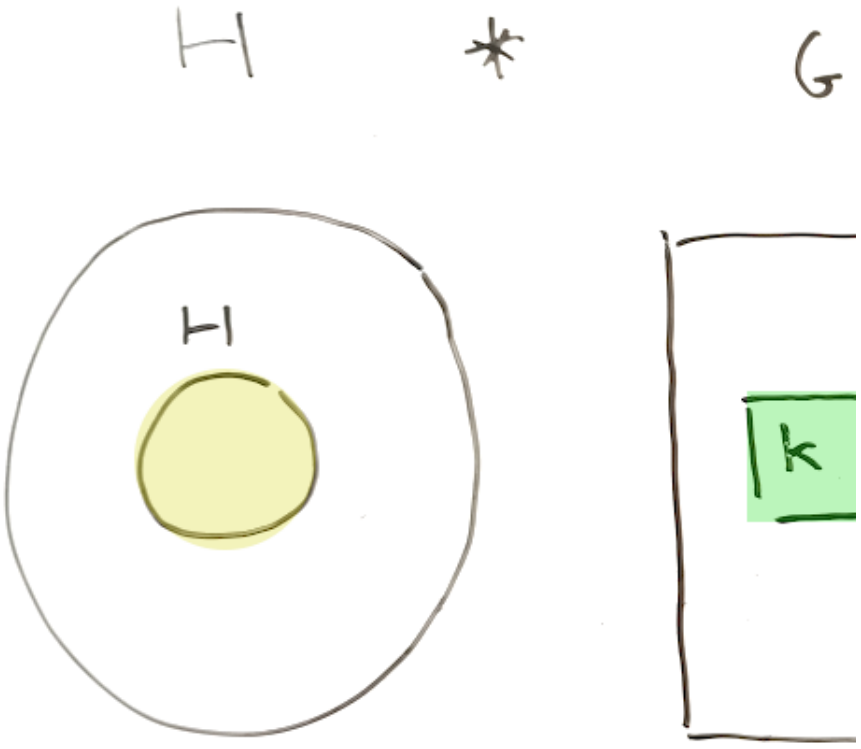
$K_a = H \cap K^a$



$H \mid K_b$

$K_b = H \cap K^b$





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== Conjugation ==

Conjugacy class: https://en.wikipedia.org/wiki/Conjugacy_class

Equivalence	**Class Set**	
cosets aH	**aH** a ∈ G G → **G|H** |aH| = |H|	**H** aH =
bH <=> a ~ b mod H	a ↦ aH	
	ker = **H** aH **partitions** G	
	# of distinct aH (classes) = |G|/|H|	
	class size = |H|	
conjugate S^a	**S^a** a ∈ G G → { S^a | a ∈ G } ≃ **G|N(S)** |S^a| = |S|	**N(S)** S^a =
S^b <=> a ~ b mod N(S)	a ↦ S^a ≃ N(S)	

ker = **N(S)**
neither partitions nor disjoint
e.g. S^a = S for ∀a if S is normal
of distinct S^a (classes) = |G|/|N(S)|

class size = |N(S)|

inner bijection s ↦ s^n

φ**n**: S → S ∈ Bij(S) n ∈ N(S) **C(S)**
N(S) → Inn(S) ≃ **N(S)/C(S)** note: C(S) is normal

φn(s) = s^n φn = φm <=> n ~ m mod C(S)
n ↦ φn

ker = **C(S)**

of distinct φn (classes) = |N(S)|/|C(S)|

class size = |C(S)|

conjugacy class a^G

a^G a ∈ G a^G = b^G <=> b ∈ a^G

|a^G| = |G|/|C(a)|

Idea: conjugation can be interpreted as a measure of **commutativity**
i.e. how close an element is to being commutative

a^G = {a} <=> a is commutative

centralizer C(a) = G

Conjugation
Let G be a group.

Conjugate of an element a by g is

a^g := gag^-1 for some g ∈ G

The set of all conjugates of a is its **conjugacy class**

a^G = { a^G| g ∈ G }

Conversely, the **conjugate of a set S by g** is the set

S^g := { a^G| a ∈ S }

S^G := { S^g: g ∈ G }

A subgroup N ⊆ G **invariant** under all conjugation is called a **normal**

subgroup
i.e.

N^g = N for all g ∈ G note: this is the same as N^g ⊆ N since |N^g| = |N|

G is **simple** if the only normal subgroups are **trivial** i.e. {1} and G.

Theorem: basic properties
Let S ⊆ G, H ≤ G

associativity

a) **element** conjugate

b) **set** conjugate

identities

a) **element** conjugate

a^g = b^g <=> a = b **uniqueness**

a^g = a^g' <=> g' ∈ gC(a) i.e. each class is characterized by the **centralizer**

a^G ≃ G|C(a)

g ↦ a^g g ↦ gC(a)**element**

|a^G| = |G| / |C(a)|
order

a^g = b^g' <=> b ∈ a^G i.e. conjugacy classes are **partitions**

a^G ∩ b^G = a^G or ∅

associativity
(ab)^g = a^g b^g
a^gh = (a^g)^h

b) **set** conjugate

S^g = S^g' <=> g' ∈ gN(S) i.e. each conjugate is characterized by the normalizer

S^G ≃ G|N(S)

g ↦ S^g g ↦ gN(S) **element**

|S^g| = |S|

order

|S^G| = |G| / |

N(S)|

associativity

ST^g = S^g T^g
S^(gh) = (S^g)^h

However, sets do **NOT** form a partition.
Indeed, any commutative x ∈ S are in all S^g

ZG ∩ S ⊆ Core (S)
i.e. in all S^g

In general, the intersection is determined by H ∩ H^x
i.e. for any b = ax

H^a ∩ **H^b** = (H ∩ H^x)^a

partition a^g conjugacy classes a^G form a **partition**
conjugacy class Let S ⊆ G be **closed** under conjugation
i.e. S^G = S
Then

$$S = \biguplus_{a \in S} a^G$$

a^G ∩ b^G = a^G or ∅ by the partition identity a^g = b^g' <=> b

Each class, in term, is determined by its centralizer

set	a^G	≃	G|N(a)
class	g ↦ a^g	<=>	g ↦ gN(a)

with size

|a^G| = |G| / |N(a)|

Class equation

$$|S| = \sum_{\text{over } a^G} |a^G|$$

$$= \sum_{\text{over } a^G} |G| / |N(a)|$$

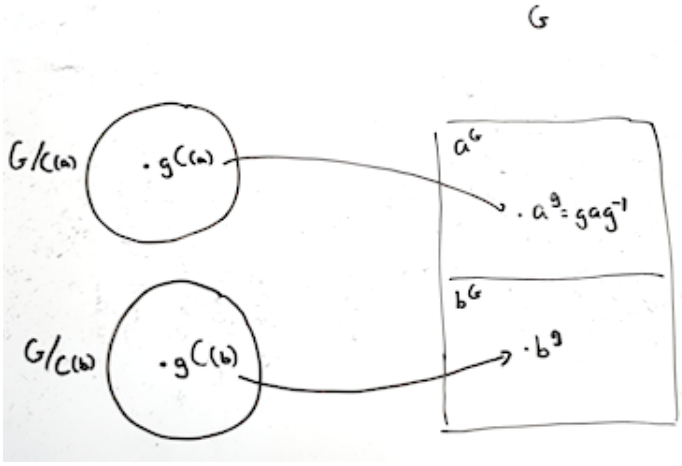
$$= \sum_{\text{over } N(s) = G} 1 + \sum_{\text{over } N(a) \neq G} |G| / |N(a)|$$

i.e. a **commutes**

$$= |ZG \cap S| + \sum_{\text{over } N(a) \neq G} |G| / |N(a)|$$

In particular, if S = G

$$|G| = |ZG| + \sum_{\text{over } N(a) \neq G} |G| / |N(a)|$$



group H^g	Unlike conjugate each H^g is a **subgroup** TODO

Inner automorphism
https://en.wikipedia.org/wiki/Inner_automorphism

Actions by g on N are **bijection** (aka permutation, symmetry N)

φg: N → N
φg(a) = a^g

In particular, if N is a subgroup, then φg is a **group homomorphism** i.e. an **automorphism of N**

product preservation

φg(ab)	=	(ab)^g
	=	g(ab)g^-1
	=	(gag^-1)(gbg^-1)
	=	φg(a) φg(b)

φg is called an **inner automorphism of N**.

Furthermore, if N = G, then the set of all inner automorphisms is the **inner automorphism group Inn(G) ⊆ Aut(G)**

Inn(G) = { φg: G → G| g ∈ G }

φg ∘ φh	= φgh	group operation
(φg)^-1	= φ(g^-1)	inverse

Inn(G) is **normal** in Aut(G), so the quotient exists.
This is the **outer automorphism group Out(G)**

Out(G) = Aut(G) / Inn(G) note: Inn(G) is normal since f∘φg∘f^-1 =φ(f(g)) ∈ Inn(G) for any f ∈ Aut(G)

==========

== Normalizer, Centralizer, Core ==

Normalizer N(S)
Centralizer C(S)
Let G be a group and S ⊆ G.
Then

normalizer N(S) := { g ∈ G| gS = Sg }
= { g ∈ G| S^g = S }
i.e. all g **commuting** with S **set-wise**

core Core(S) := ∩ S^g
g ∈ G

center C(S) := { g∈ G | gs = sg for all s ∈ S }
= { g ∈ G | s^g = s for all s ∈ S }
i.e. all g **commuting** with S **element-wise**

= ∩ C(s)
s ∈ S

= { g ∈ G | S ⊆ C(g) }

Special cases

S = G	**C(G) = ZG**	i.e. center of G
	G = Core(G)	trivial
	G = N(G)	
S = point a	C(a) = N(a)	

N(S) and C(S) are **subgroups**
e.g. for any n ∈ N(S)

inverse	S^(n^-1)	= (S^n)^(n^-1)	by normality
		= S	
operation	S^(nm)	= (S^n)^m	by normality
		= S^m	
		= S	

Core(H) is a **subgroup** for H subgroup since each H^g is a subgroup

inverse x ∈ H^g <=> x^-1 ∈ H^g for all x

If H is a subgroup, then H is always a normal subgroup of N(H).

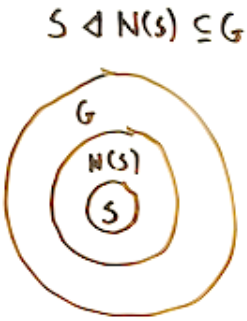
H ◁ N(H)

In fact, it is the **largest** subgroup of G in which H is normal

H ◁ K in G => **K ⊆ N(H)**

In particular,

H ◁ G <=> G = N(H)



Theorem: basic properties
Let G be a group and S ⊆ G, H ≤ G
Let

N^i(S) := N(N(...N(S)) i times
C^i(S) := C(C(...C(S))

foundation

N(S), C(S) are all **subgroups**

H is normal <=> G = N(H)

a ∈ G commutes <=> C(a) = G

H is abelian <=> H ⊆ C(H)

H ◁ N(H) N(H) is **largest** normal subgroup of S in G
i.e. for any subgroup K of G

H ◁ K => K ⊆ N(H)

C(S) ◁ N(S) C(S) is exactly the **kernel** of fixed conjugations

i.e.
C(S) = ker [^]

N(S) / C(S) ≃ Inn(S)

nC(S) n^: S → S

conjugation

N(S^a) = N(S)^a

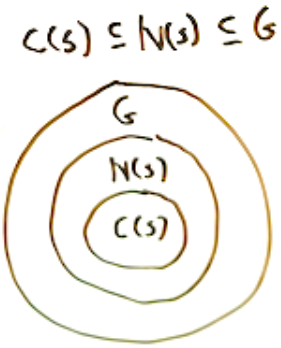
C(S^a) = C(S)^a

inclusions

inclusions of N(S)

N(S) contains both S and C(S)

$S \subseteq N(S)$
$C(S) \subseteq N(S)$



normal series

subgroup		**quotient**
H	$\lhd$ N(H)	N(H) / H
N(H)	$\lhd$ N^2(H)	N^2(H) / N(H)
N^2(H)	$\lhd$ N^3(H)	N^3(H) / N^2(H)
....		

inclusions of C(S)
On the other hand, C(S) does NOT necessarily contain S

$H \subseteq C(H)$ <=> H is abelian

Instead there are **symmetries**

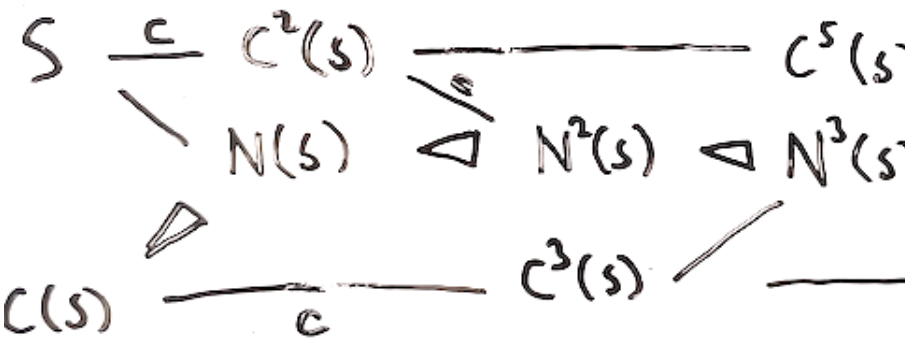
$T \subseteq C(S)$ <=> $S \subseteq C(T)$

Hence

$S \subseteq C(C(S))$ inclusion
$C(S) = \{ g \in G \mid S \subseteq C(g) \}$ another definition of C(S)
since $g \in C(S)$ <=> $S \subseteq C(G)$

$S \subseteq C^2(S) \subseteq C^4(S) \ldots$ hierarchy
$C(S) \subseteq C^3(S) \subseteq C^5(S) \ldots$

$C^i(S) \subseteq N^i(S)$ note: since $C(C(T)) \subseteq N(C(T)) \subseteq N(N(T))$ for any T



Normalizer & center

The center and core both occur as kernels of an action

action	kernel
inner automorphism G / C(G) ≃ Inn(G)	ker ^g = C(g)
^ : G x G → G x ^g := x^g	ker [^: G → Aut(G)] = C(G) = G / ZG
inner bijections of X ⊆ G N(X) / C(X) ≃ Inn(X)	ker ^n = C(n) ∩ X
^ : X x N(X) → X x ^n := x^n	ker [^: N(X) → S(X)] = C(X)
coset * : G x G|H → G|H g * aH = (ga)H	ker * aH = H^a ker [*: G → S(G|H)] = Core(H) = ∩ H^a

A special case is when S = G

Z(G) = C(G)	i.e. center
N(G) / Z(G) ≃ Inn(G)	where Inn(G) ⊆ Aut(G) is the subgroup of inner automorphisms

==========

proof

- C(S) ◁ N(S)

proof 1: **indirect**

C(S) is the kernel of the map

φ: N(S) → Sym(S)	
φg: S → S	inner bijection φg(s) = gsg^-1 ∈ S

and a kernel is normal.

proof 2: **direct**
Pick any $c \in C(S)$, $n \in N(S)$ and $s \in S$

$c^n \; s = (ncn^{-1}) \; s \; nn^{-1}$
$= n \; c \; (n^{-1}sn) \; n^{-1}$
$= n \; (n^{-1}sn) \; c \; n^{-1}$ since $n^{-1}sn \in S$ so it commutes with c
$= s \; c^n$

$\Rightarrow ncn^{-1} \in C(S)$

QED
==========

Normalizer & core
TODO
The core is

$C(S) \subseteq Core(S)$

$C(S) \subseteq$

Note that

Class Equation
Idea: TODO

Theorem: class equation
Let G be a group

a) **element** class equation
Let action

$*: H \times G \to G$
$h * a = a^h$

Then

$|G| = \sum_{\text{over } a^H} |a^H|$

$= \sum_{\text{over } a^H} |H| \; / \; |N(a)|$

$= \sum_{\text{over } N(a) = H} 1 \; + \; |H| \sum_{\text{over } a^H \text{ with } N(a) \neq H} 1 \; / \; |N(a)|$

In particular, if H = G then

$|G| = |ZG| + |G| \sum_{\text{over } a^H \text{ with } N(a) \neq H} 1 \; / \; |N(a)|$

b) **subgroup** class equation
Let action

*: H x G|K → G|K
h * aK = haK

Then

|G|K| = ∑ | H * aK|
over aK

= ∑ |H| / |H ∩ K^a|
note: since kernel *aK = H ∩ K^a

= ∑ 1 + ∑ |H| / |H∩ K^a|
over H ⊆ K^a rest

In particular, if H = K then

|G|H| = ∑ 1 + ∑ |H| / |H ∩ H^a|
aH with H = H^a aH with a ∉ N(H)
i.e. a ∈ N(H)

= |N(H) / H| + ∑ |H| / |H ∩ H^a|
aH with a ∉ N(H)

= [N(H):H] + ∑ |H| / |H ∩ H^a|
aH with a ∉ N(H)

==========
====
== Normality ==

Normal subgroup: https://en.wikipedia.org/wiki/Normal_subgroup

Normal subgroup

NOTE: **normality** is a type of **invariance** under a translation (conjugation)

N^g = N for all g ∈ G

Normality can also be understood as a **weak commutativity**

gN = Ng f or all g

and hence is a sufficient in some important cases.
Normality is **canonical** in that it is the critical requirement for **quotients**.

N is **normal** <=> **quotient G/N** is well-defined

Theorem: normality
Let N be a normal subgroup.

a) **conjugacy class**: N contains all conjugacy classes of its elements.
i.e.

$n^G \subseteq N$ for all $n \in N$

Hence if N is a subgroup

N is normal <=> **$N = \cup n^G$** over all $n \in N$

<=> $N = N^g$ for all $g \in G$

b) **weak abelian**:

$gN = Ng$ for any $g \in G$

i.e. for any $n \in N$

$gn = n'g$ for some $n' \in N$

c) **quotient**: the quotient G/N exists
In particular,

$aN \cdot bN = (ab)N$ since aNbN = a(bN)N = abN by normality

is well-defined and

$|G/N| = [G:N]$
$= |G| / |N|$

d) **lattice**: normal subgroups form a lattice
Let H, N be subgroups of G
Note that (see products section)

N or H normal => **$H \cap N$ normal**
N and H normal => **HN = NH normal**

Hence

$H \cap N$ <= H, N <= **HN** i.e. $H \cap N$ is the **largest** normal subgroup of $G \supseteq H,N$
greatest lower least upper HN is the **smallest** normal subgroup of $G \supseteq H, N$
bound bound

1-1 relation (aka isomorphism theorem):
If N is normal in G, then

$H \cap N$ normal in H

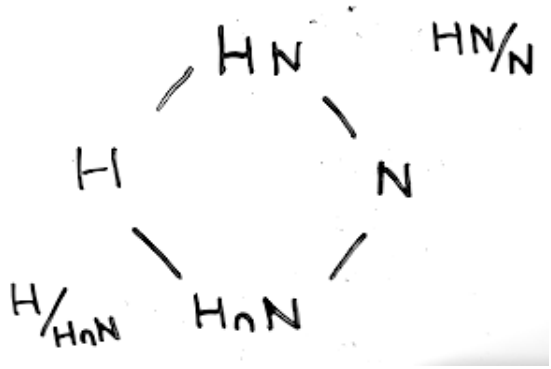
and

$HN / N \simeq H / H \cap N$ since for any $h,h' \in H$
$hN = h'N$ <=> $h = h'n$ for some $n \in N$

note:

hence $n \in H \cap N$

<=> $h(H \cap N) = h'(H \cap N)$



e) **refinement**:
Let H, H' be subgroups and N ◁ H, N' ◁ H'
Then there is a refinement of H ∩ H' ⊆ H
i.e. an intermediate subgroup **(H ∩ H')N** with

H ∩ H'	⊆	**(H ∩ H') N**	⊆	H
◁		◁		◁
(H ∩ N)(H ∩ N')		**(H ∩ N') N**		N

note: (H ∩ H)N, (H ∩ H')N are subgroups by normality of N,

related by

(H ∩ H')N / (H ∩ N')N ≃ (H ∩ H')(H' ∩ N) / (H ∩ N')(H' ∩ N)

=

(H ∩ H') / (H' ∩ N) (H ∩ N')

since H' ∩ N ⊆ H ∩ H'

=

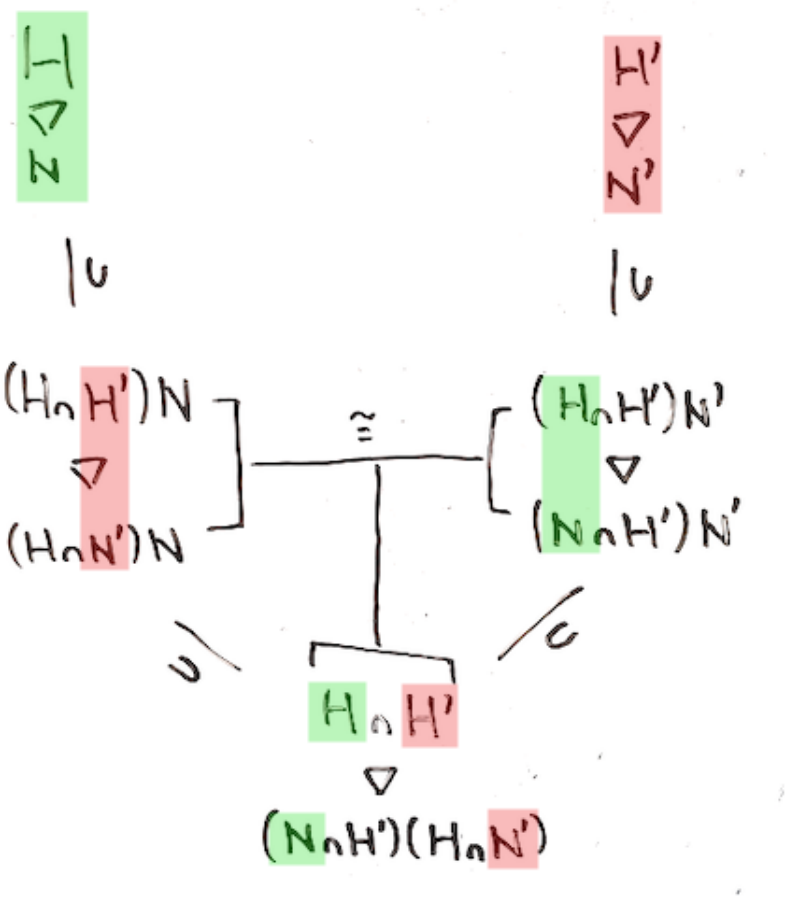
(H ∩ H')(H∩ N') / (H' ∩ N)(H ∩ N')

≃ **(H' ∩ H)N' / (H' ∩ N)N'**

i.e.

AN/BN ≃ **AN'/CN'** ≃ **A/BC**

where A = H ∩ H'
B = H ∩ N'
C = N ∩ H'



==========

Proof:

b) Pick any n ∈ N. There exists some n' such that

n = g^-1n'g since N^(g^-1) = N

=> gn = g(g^-1 n' g)
= n' g

c)

Let a, b ∈ G

Need to verify that aN bN = (ab)N is **well-defined**.

i.e. that

(ab) ~ (a'b') mod N for any aN = a'N, bN = b'N

This is because

a'b' = (an)(bn') for some a' = an, b' = bn'
= a(bn'')n' nb = bn'' for some n'' by normality
= abn''n' ∈ (ab)N

=> a'b' ~ ab mod N

QED

==========

In fact, these are all equivalent formulations of normality.

Theorem: definitions

Let N be a subgroup of G.
The following are equivalent formulations of a normal subgroup

a) **invariant**: N^g = N for all g ∈ G
note: this is the same as N^g ⊆ N

b) **weak abelian**: left and right cosets coincide
i.e.

gN = Ng for any g ∈ G

c) **conjugacy classes**: N is a union of conjugacy classes
i.e.

N = ∪ n^G

d) the **quotient group G/N** exists
note: normality needed to define aN * bN = (ab)N

e) **kernel**: there is a group homomorphism ϕ: G → H such that ker ϕ = N
i.e. the quotient map ϕ: G → G/N

f) **extension (g)N**: for every g ∈ G, the product

(g)N

is a subgroup.

==========
Proof:

e)

<=) Suppose G/N exists
Then for any g ∈ G,

gN · g^-1N = N
=> gng^-1 ∈ N for any n ∈ N
i.e. N^g ⊆ N

QED
==========

Some large subgroups are forced to be normal.

Theorem:
If there is a subgroup H of G with

[G:H] = p
where p = **smallest** prime divisor of |G|

then H is normal.
In particular, if

[G:H] = 2

then H is normal.

==========
Proof 1:
idea: since p is already maximal,

Consider how each $g \in G$ acts on the p cosets of H

$g \in G \mapsto$ (gH, ga1H, ..., ga(p-1)H)

Each g permutes the cosets, so this is a group homomorphism

$G \to Sp$

=> if K = kernel, then |G/K| | |Sp| = p!
=> |G/K| = p since p is the smallest divisor

On the other hand, $K \subseteq H$ since gH = H for every $g \in K$

```
=> p    = |G/K|
          = [G:K]
          = [G:H][H:K]
          = p[H:K]

=> [H:K] = 1
   i.e. H=K
=> H is normal since kernels are normal
```

Proof 2:
Idea: consider actions of H on G|H
Let

$*: H \times G|H \to G|H \simeq Cp$
$h * aH = h(aH)$

Then

$$p = \sum |H * aH|$$
$$= \sum |H| / |H \cap H^a| \quad \text{since ker } *aH = H^a$$
$$= 1 + \sum_{a \neq 1} |H| / |H \cap H^a|$$

=> all prime factors of |H| are >= p so it must be that
$|H \cap H^a| = |H|$
=> $H \cap H^a = H$ for all $a \in G$ since it is independent of the choice of representative.

QED
==========

Series
Normal, composition series
https://en.wikipedia.org/wiki/Subgroup_series
https://en.wikipedia.org/wiki/Composition_series

Let G be a group.
If H0, ..., Hn are a sequence of subgroups, then

normal series subgroups Hi i = 0..., n where

$H0 = \{1\}$ $Hn = G$

$H0 \subseteq H1 \subseteq H2 ... \subseteq Hn$

$Hi\text{-}1 \lhd Hi$

factor group quotient Hi / Hi-1

length |Hi| = n if each Hi is a **proper** subgroup
i.e. $Hi+1 \neq Hi$

A series H'i is a **refinement** of Hi if H'i contains Hi
A series is a **composition series** if it is **maximal**.
i.e. has no **proper** refinement.

A composition series Hi provides a **decomposition** of G into **factor** products
i.e.

group H1 x H2/H1 x H3/H2 ... G

element (a1, a2H1, a3H2, ...) <=> $g \in G$
...a3 · a2 · a1 = g

Solvable group
A group G is **solvable** if the factor groups of the composition are all **cyclic**.
This implies that the group is constructed by appending a single element
i.e.

$H0 = \{ 1 \} \lhd H1 \lhd H2 \lhd H3$ ...
$H1/H0 \simeq (a0)$ $H2/H1 \simeq (a1)$ $H3/H2 \simeq (a2)$

Hence, for every i

$Hi+1 = (ai)Hi$
$= Hi(ai)$ by normality

In particular

G = ...(a3)(a2)(a1)
= (a1)(a2)(a3)... by normality
NOTE: order matters though
i.e. either ascending or descending

Theorem: composition series

refinement

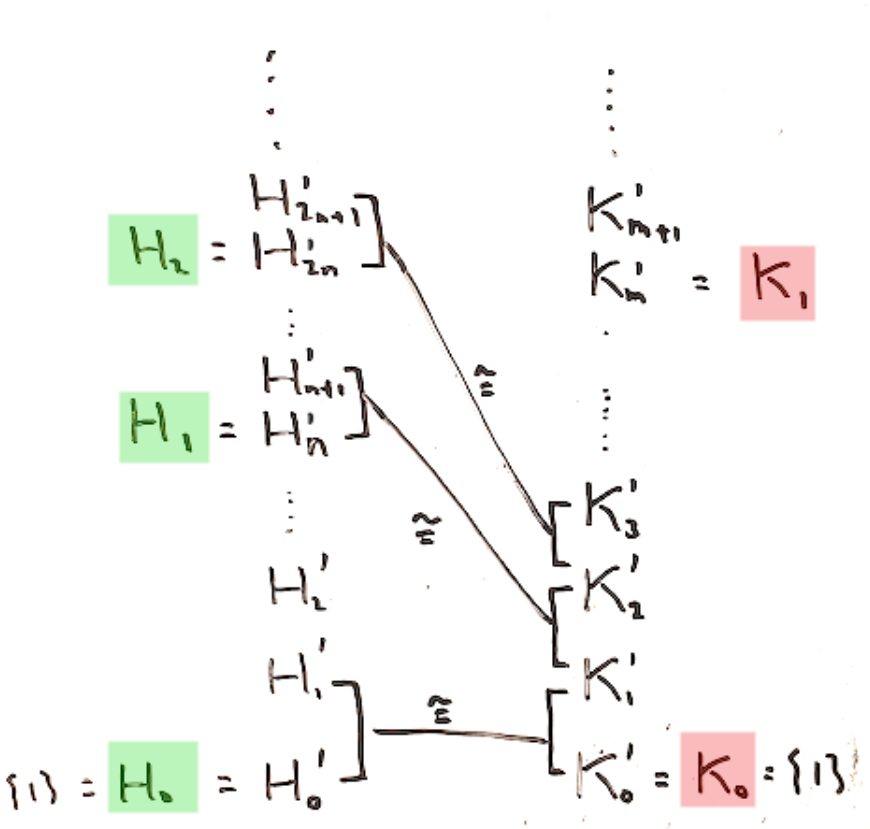
Two normal series can be **combined** up to **isomorphism**
i.e.

normal series Hi => **refinements** H'i of Hi of the same **length**
Kj K'i of Ki

with 1-1 relationship between factor groups.
Unique pairing **(Hi, Kj)** such that

H'i+1 / H'i ≃ K'j+1/Kj note: NOT necessarily i = j

for every Hi and Kj.



uniqueness

Composition series is **unique** upto **isomorphism**.
i.e. for any composition series Hi, Ki

factor groups **isomorphic** pairing of factor groups Hi+1/Hi ≃ Kj+1/Kj

Each group is **simple**

length |Hi| = |Ki|

In fact

Hi composition series <=> every factor group is **simple**

note: by 1-1 of normal subgroups of Hi+1 and its factor

group Hi+1

Hi+1/Hi

subgroup S ◁ Hi+1

<=> S' ◁ Hi+1/Hi

S = SHi **S = S'Hi**

so Hi has a refinement iff

Hi+1/Hi NOT simple.

The **composition length** of a group G then is

c(G) := length of composition series

decomposition

For any H normal subgroup with

H0 ◁ ... ◁ Hn composition series of H

K0/H ◁ .... ◁ Km/H composition series of G/H

then

H0 = {1} ◁ H1 ... Hn-1 ◁ Hn = K0H ◁ ... ◁ KmH

note: by 1-1 correspondence of quotients

(Ki+1/H) / (Ki/H) ≃ Ki+1H /KiH

i.e. (Ki+1/H) / (Ki/H) simple iff Ki+1H /KiH simple

is a composition series of G and

c(G) = c(H) + c(G/H)

abelian

If G is abelian, then it is solvable and

c(G) = ∑ n_p i.e. over all prime divisor of |G|
p^n_p || |G|

==========

proof

- series:
Idea: **subdivide** the interval each series by the other

series K0 K1 K2
K1 ∩ H0, K1 ∩ H1, ... K2 ∩ H0, K2 ∩ H1, ...

Consider the sequence
K0 = H0 = {1}
Km = Hn = G

Hn-1(Hn ∩ Km) ≃ Hn = G
Hn-1(Hn ∩ Km-1)

...

H1(H2 ∩ K1)
H1(**H2** ∩ **K0**) ≃ H1

H0(H1 ∩ Km) ≃ H1

....

H0(H1 ∩ **K2**)
H0(H1 ∩ **K1**)
H0(H1 ∩ **K0**) ≃ H0 **=** {1}

Similarly,

Km-1 (Km ∩ Hn) ≃ Km **=** G
Km-1 (Km ∩ Hn-1)

...

K1 (K2 ∩ **H1**)
K1 (K2 ∩ **H0)** ≃ **K1**
K0 (K1 ∩ **Hn**) ≃ K1

....

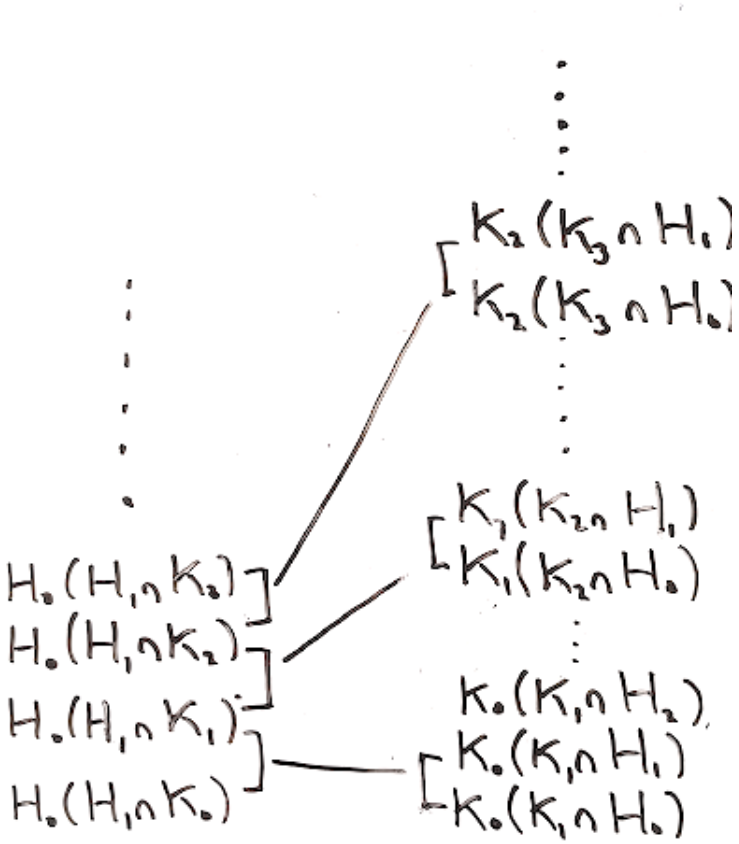
K0 (K1 ∩ **H2**)
K0 (K1 ∩ **H1**)
K0 (K1 ∩ **H0**) ≃ K0 **=** {1}

Then there is a 1-1 correspondence between the sequences

Kj (Kj+1 ∩ Hi+1) / Kj (Kj+1 ∩ Hi) ≃ (Hi+1 ∩ Kj+1) / (Kj+1 ∩ Hi) (Hi+1 ∩ Kj)

≃ Hi(Hi+1 ∩ Kj+1) / Hi(Hi+1 ∩ Kj)

by the composition isomorphism



- composition series

QED
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Characteristic
https://en.wikipedia.org/wiki/Characteristic_subgroup

Characteristic is a stronger version of normality.
If H is **normal** then H is closed under the **conjugation automorphism**.
i.e.

φ**(H) = H^g = H** for any φ = ^g $\in$ Inn(G)

H is **characteristic** if

φ**(H) = H** for any $\varphi \in$ **Aut(G)**

Bky definition, characteristic subgroups are normal.

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== Quotient group ==

Quotient group: https://en.wikipedia.org/wiki/Quotient_group

Quotient subgroup

Let G/N be a quotient.
A subgroup of G/N can be written as

H'/N = { hN: h ∈ H' } H' ⊆ G

where H' is only a set of **representatives**, not necessarily a subgroup.
But there is a 1-1 relationship

G		**G/N**
subgroup H = HN	<=>	subgroup H'/N
H = H'N		

so we can always assume that H' is a subgroup H' = H'N.
Hence, we can write

(H/N) N = H if subgroup H = HN

Theorem (basic)
Let G be a group, N a normal subgroup with quotient map π: G → G/N

a) **subgroups** of G/N
https://en.wikipedia.org/wiki/Correspondence_theorem

There is a 1-1 correspondence (aka **isomorphism/correspondence** theorem):

G		**G/N**
subgroup **H = HN**	<=>	subgroup H/N
subgroup H where N ⊆ H	<=>	

Specifics

	G		**G/H**
subgroup of G	subgroup H	=>	subgroup π(K) = H/N
	subgroup π^-1(π(H)) = HN	<=	
	Hence **H = HN** iff K = π^-1(π(H))		
subgroup of G/H		<=	subgroup H'/N
	subgroup π^-1(H') = H'N		
		=>	subgroup π(H'N) = H'N/N = H'/N
normality	normality is **preserved**		
	H is **normal**	=>	H/N is normal

H'N is normal <= H'/N is normal

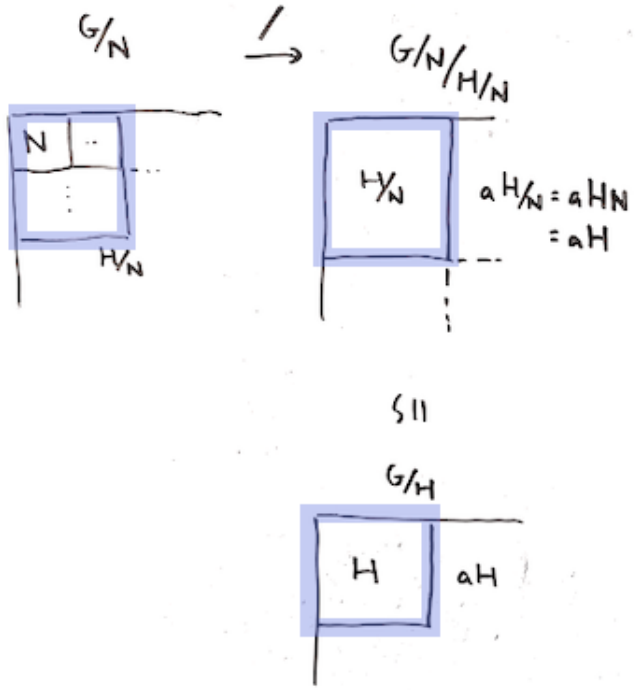
note:
H is normal => HN is normal

quotient Hence **quotients** are also preserved.
If $H \lhd G$

$G / H \simeq (G/N) / (H/N)$

gH <=> gN(HN) = gH

since NH = HN by normality



cyclic subgroup (a) of order n => (a)/N of order n

$(a) \cap N = \{1\}$

(a') of order nk <= (a')/N of order n in a **finite** G

note: by Fermat's little theorem
for some k
$a'^n \in N$ so $a'^{(n|N|)} = 1$

==========

Proof

- composition:

- $H \cap N' \lhd H \cap H'$: for any $h \in H \cap H'$ and $n \in H \cap N'$

hn = n'h for some $n' \in N'$

=> $n' \in H$ since h, n are both in H

i.e. $n' \in H \cap N'$

- $N(H \cap N') \lhd N(H \cap H')$

$N(H \cap H')\, N(H \cap N')$	=	$N\,(N\,(H \cap H'))\,(H \cap N')$	by normality of N
	=	$N\,(H \cap H')\,(H \cap N')$	
	=	$N\ \ (H \cap N')\,(H \cap H')$	by normality of N'
	=	$N\ \ ((H \cap N')\,N)\,(H \cap H')$	by normality of N
	=	$N\,(H \cap N')\ \ N(H \cap H')$	i.e. commute

=> $N(H \cap N')$ is normal in $N(H \cap H')$

QED

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== Malnormality ==

Malnormal subgroup

A **malnormal** subgroup H is a subgroup where all conjugates of H are distinct.

i.e.

$H^a \cap H^b$	=	H^a	if $b \in aH$
		1	otherwise

As a special case with b = 1,

$H^a \cap H = 1$ for all a	=>	$N(H) = H$ i.e. **self normal**

Hence, malnormality is **stronger** than self-normal.

Frobenius group

https://en.wikipedia.org/wiki/Frobenius_group

Let H be malnormal and K be the set of all **non-conjugates** of H

$K := 1 \cup$ **(**$G \setminus \cup H^a$)

i.e. all k NOT a **conjugate**

K is called a **Frobenius kernel**.

Then it turns out K is normal

K is **normal** since if

$K^g \cap H^a \neq \{ 1 \}$

=> $K \cap H^{(ga)} \neq \{ 1 \}$

K	=	$\|G\|$	- $\sum (\|H^a\| - 1)$ over unique H^a
	=	$\|G\|$	- $[G:H]\,(\|H\| - 1)$
	=	$[G:H]$	

G can be **decomposed** into a semi-direct product $K \rtimes H$
G is called a Frobenius group

G	$\simeq$	K	$\rtimes$	H
Frobenius group		**Frobenius Kernel**		**Frobenius Complement**

Theorem: definition
Let H be a subgroup of G
Let

*: G x G|H → G|H
g * aH = (ga)H

Equivalent definitions of malnormality
a) **partition**

H^a ∩ H = 1 for all a ∉ H

b) **partition**

H^a ∩ H^b =	H^a	if b ∈ aH
	1	otherwise

c) **coset**

aH ∩ Ha = { a } for all a ∉ H

d) **fixed point**: each permutation πg: G|H → G|H has at most one **fixed point**
i.e.

g ∈ G^a	<=>	πg(xH) = xH	only for x = a
k ∈ K	<=>	πk(xH) ≠ xH	for all x

e) **fixed point**: let SX be a group of permutations where each permutation has at most one fixed point.
Let i0 ∈ X be a fixed point and

H = { g(i0) = i0 | g ∈ SX }
i.e. permutations

Then the permutation group of S|H is isomorphic to SX

group	S(S|H)	$\simeq$	SX
element	g ∈ aH	<=> a(i0) = a0	g(i0) = a0
	πg(a) = b		ga(i0) = b(i0)
fixed point	πg(a) = a		g(a0) = a0

Thus, S|H provides a **representation** of the permutations in SX without knowing any of the actual permutation of each g ∈ SX except g(i0).

f) **image**:

|H * aH| = |H| note: |H * aH| = |H| / |H^a ∩ H| so
|H * aH| = |H| <=> |H^a ∩ H| = 1

Hence the permutations πh: G|H → G|H are **unique** on all values except 1
i.e. for any h, h' ∈ H

πh(aH) ≠ πh'(aH) ≠ aH for all a ∉ H

Theorem: Malnormal
Properties

Kernel K

k^g ∈ K for any k ∈ K

coset aH

|aH ∩ H^g| = 1 or 0 for all g
i.e. aH has **at most one h^g**

since
h^g, h'^g ∈ aH
=> h^g = h'^g h" for some h" ∈ H
=> (h'^-1h)^g = h"
=> h" =1, h = h' since H^g ∩ H = 1

|aH ∩ K| = 1

Hence aH is composed of single items from both K and H^g

aH single k ∈ K

at most single h^a ∈ H^g for some g ≠ a

Order of [G:H]

[G:H] = |G| / |H|

Also, counting unique cosets

[G:H] = # of cosets

= ∑ |H(x)| where H(x) = { hxH | h ∈ H }
over all unique image i.e. # of unique cosets hxH h

= 1 + ∑ |H| since |H(1)| = 1 fixed point x = a

x = 1 over x ≠ a |H(x)| = |H|
otherwise by uniqueness (see below)

Hence with n = |H|

n | |G|

n | [G:H] -1
= (|G| / n) - 1

|G| = n(kn + 1) for some k

symmetry

For simplicity, write πg: G|H → G|H

πg(a) = b <=> g * aH = gaH = bH

Then

conjugate g ∈ H^a
fixed point πg(a) = a
πg(x) = a x ∈ aH

uniqueness πg(x) ≠ πg'(x) x ∉ aH, distinct g' ∈ H^a

note: similar reasoning as coset above

partition Each image H^a * xH^a partitions the permutations
i.e.

πg: (H^a * xH) → (H^a * xH)
πg(πg'(x)) = gg' * x

complement K
πk(a) ≠ k for all a <=> k ∈ K

coset g ∈ aH
identity πg(1) = a
πg(x) = a x ∈ H

uniqueness πg(x) ≠ πg'(x) x ∉ H, distinct g' ∈ aH

note: since if πg(x) = πg'(x)
=> ahxH = ah'xH for g = ah, g= ah'
=> (h^-1h')^x ∈ H
=> x ∈ H since H^x ∩ H = 1

==========
proof

Let g ∈ G and g(a) = b
For any h ≠ 1

g^-1 h^b g(a) = a

=> h^(g^-1 b) ∈ H^a

=> It must be that g^-1 b ∈ aH since otherwise,

h^(g^-1 b) = 1 since H^(g^-1 b) ∩ H^a = 1

=> h = 1 contradiction

=> g^-1b ∈ aH

Hence g ∈ bHa^-1

QED
==========

TODO
==========
proof of Frobenius

Main observation
For any g, g' ∈ G

πg(a) = b	<=>	gg' ∈ H^a
πg'(b) = a		g'g ∈ H^b
	<=>	gh^bg' ∈ H^a for any h
		g'h^ag ∈ H^b

So want to prove that

πk(a) ≠ πk'(a) for all k, k' ∈ K

Observation
Equivalent

a) K is a subgroup

b) πk(a) ≠ πk'(a) for all distinct k, k' ∈ K

c) |aH ∩ K| = 1 for every a ∈ G
i.e. each aH has exactly one
k ∈ K

proof of c <=> a:
Since

k', k ∈ aH
<=> kH = k'H
<=> kk'^-1 ∈ H

Observation
If k'k ∈ H^a, then

k'h^bk, k^-1h^bk ∈ H^a for all h

H^a = { k'h^bk | h ∈ H }
= { k^-1'h^bk | h ∈ H }

=> k'hk = k^-1h'k for some h, h'
=> k' ∈ k^-1H

Observation: cosets kH
If K is a subgroup, then it must be that cosets G|H are **exactly** all the cosets kH
i.e.

unique kH ≠ k'H if k ≠ k' since kH = k'H => kk'^-1 ∈ H

all G|H = { kH | k ∈ K }

Hence

kh ∈ H^a for some a

i.e. for each h ∈ H except h = 1, there is a unique (a, b)

πk(a) = b πh(b) = a note: otherwise kh ∈ K

How to prove??

NOTE: If for some x

y = πk(x) ∈ H * xH

=> there exists h ∈ H such that πh(y) = x
=> πkh(x) = x
i.e. kh ∈ H^x

Hence, there must be **AT MOST |H|** values of x such that

πk(x) ∈ H * xH

since otherwise by pigeon hole there is at least one k such that

πk(x) ∈ H * xH πk(x') ∈ H * x'H

=> kh ∈ H^x ∩ H^x'
contradiction

Q: why?

Observation: conjugation k^g
If k ∈ K

k^g ∈ K for any g ∈ G

Hence

k ∈ aH => hkh^-1 ∈ haH
i.e. every coset in H * aH has an element of K

QED
==========

Transfer map
Transfer map
https://en.wikipedia.org/wiki/Transfer_(group_theory)
Let ai be representatives of H cosets.
Then for any g ∈ G and ai,

g · aiH = ai'H
g · ai = ai' · h_g,i for some h_g,i ∈ H

The **transfer map** is

V: G → H/H' where H' = [H,H]
V(g) = ∏ h_g,i mod H'

This is **independent** of the choice of representatives.
i.e. If bi = ai · hi is another set of representatives, then

g · bi = g · (ai · hi)
= (ai' · h_g,i) · hi by definition of g · ai
= bi' · (hi'^-1 · h_g,i · hi)

=> ∏ (hi'^-1 · h_g,i · hi)
= ∏ h_gi by cancellation mod H'
= V(g)

If H is **malnormal**, then

V(h) = hH'

==========
proof

- V(h) = hH' if H malnormal
Let h ∈ H and

h · ai = ai' · h_h,i

=> h_h,i = ai'^-1 · h · ai note: i ≠ i' for all ai ∉ H since H^ai ∩ H = { 1 }

Hence

V(h) = ∏ h_h,i

= ∏ ai'^-1 · h · ai

= (h'^-1 · h · h') (a^-1 · h^[G:H]-1 · a) Most ai **cancel** each other

(ai'^-1 h ai)(ai^-1 h ai'')

= ai^-1 h^2 ai''

EXCEPT

1) **last ai = a**

remaining after cancelation

2) when i = i'
i.e. when ai = h' ∈ H

= (h'^-1 • h • h') • (a^-1 • h^[G:H]-1 • a)
= **h (mod H')** = **1 (mod H')**
since H ∩ H^a = { 1 }

= h (mod H')

=> V(h) = h mod [H,H]

QED
==========

Examples

e.g. Dihedral group Dn
Reflection **subgroup H** for odd n

H = { 1, β } note: β^2 = 1

Malnormal since

x	**H^x**
rotation α^k	{ 1, α^2k β }
reflection $\alpha^k\beta$	{ 1, α^2k β }

Hence

kernel K = (α) i.e. **rotations**

G KH

= all α^k β^i for k < n, i = 0,1

e.g. Affine group of y = ax + b
Scale subgroup H

H = { y = ax | a ≠ 0 } **scale by a**

Then for any y = cx + d

(1/c x - d/c) ∘ ax ∘ (cx + d) = ax + d(a-1)/c
i.e. all lines EXCEPT a = 1

The kernel is the translations

kernel K = { x + b | b ∈ R } **translation by b**

G KH
= any ax + b for a ≠ 0, b ∈ R

==========
====

== Sylow-type Theorems ==

Theorem: prime divisor (Cauchy)
If prime p divides |G|, then G has an element of order p

==========
Proof
Idea: proof by induction on group size

claim: if G is **abelian**, then G has an element of order p
Induction on group size of G.
- Suppose |G| = p. Then G is cyclic with a p order generator
- Suppose |G| > p. Pick any element x ∈ G.
  Then
  case a) if |x| = p, then done
  case b) otherwise, by induction, G/ (x) has an element g(x) of order p
  => g^p = x^k for some k
  => 1 = (x^k)^l l = |x|
  = (g^l)^p
  => |g^l| | p
  => |g^l| = p

QED claim

For the general case, consider the class equation

|G| = |ZG| + ∑ **|G| / |Cx|** sum over unique conjugates x^G x ∉ ZG

Then

case a) if p | |ZG|, then ZG is abelian so there is an element of order p

case b) if p | |Cx| for some x, then Cx ⊆ G contains an element of order p by the induction hypothesis
note: Cx ≠ G since x ∉ ZG

case c) if p does not divide any |Cx| then

p | ∑ |G| |Cx|
p | |G|

=> p | |ZG| by class equation, which is impossible

QED
==========

Theorem: p-groups (Sylow theorems)
https://en.wikipedia.org/wiki/Sylow_theorems

Let G be a group and p^n || |G| for a prime p.

1) **p-groups (first Sylow theorem)**:
For any k <= n, G has a subgroup of order **p^k**.

It can be constructed **incrementally** by appending a new element of order p.

i.e. there exists a sequence of elements a1, ..., an ∈ G such that

order	**subgroup**		**quotient**
0	H0 = {1}		
p	H1 = (a1)H0	a1 ∈ N(H0) order p	H1 / H0 ≃ (a1) ≃ Cp
p^2	H2 = (a2)H1	a2 ∈ N(H1) order p	H2 / H1 ≃ (a2) ≃ Cp
....			
p^n	Hn = (an)Hn-1	an ∈ N(Hn-1) order p	Hn / Hn-1 ≃ (an) ≃ Cp

note: Hi-1 ◁ Hi
since Hi is a product of Hi-1 and ai ∈ N(Hi-1)

2) **p-Sylow groups (second, third Sylow theorem)**:
If p^n || |G|, then this is called a **p-Sylow group**
.
Then

H, K p-Sylow <=> K = H^a for some a ∈ G i.e. p-Sylow groups are **conjugates**

Hence if r = # of p-Sylow groups.

r = [G:N(H)] since conjugates of H ≃ G/N(H)
= [G:H] / [N(H):H]
= |G| / |N(H)|

And

r | [G:H] by [G:H] /r = [N(H):H]
= |G| / p^n

r = 1 mod p

==========
proof

1) existence
Idea: construct incrementally by taking **product** with a new element a ∈ N(H)/H

(a)H a order p
|H| = p^k

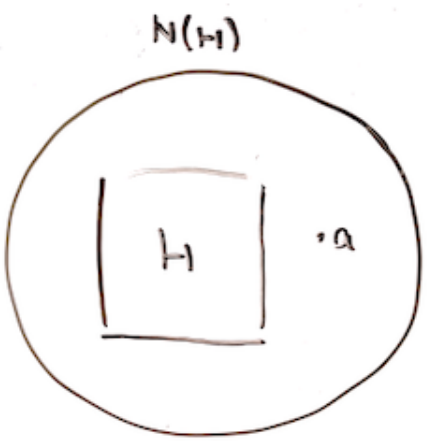
Proof by induction.
Suppose |H| = p^k for some subgroup H with k <= n.

k = 1 case: Cauchy's theorem

k < n case:
It suffices that **p^k+1 | |N(H)|**.
For if this is true, then

aH ∈ N(H)/H of **order p** by Cauchy

=> a^p ∈ H
=> there is some smallest m with a^(p^m) = 1



Let **b = a^(p^m-1)** $\in$ N(H). Then

a) b has **order p**

b) (b)H = H(b) since b $\in$ N(H) i.e. commutes with H
In particular, this is a subgroup

=> p^k = |H|
< |(b)H|
= |(b)| |H| / |(b) ∩ H | by product formula
= p^k+1 / |(b) ∩ H |

=> |(b)H| = p^k+1

and we are done.

So it remains to show that p^k+1 | |N(H)|.
Consider H ∩ H^a for all H.
i.e. class equation with K = H

|G|H| = [N(H):H] + ∑ |H| / |H ∩ H^a|
over H ∩ H^a ≠ H

=> p | |G|/|H| since k < h
p | right ∑
=> p | [N(H):H]
i.e. p^k+1 | N(H)

QED

2) classification
Idea: TODO

Let K, H be p-Sylow groups with p^n = |H| = |K|.
Consider action on G|H by K.
Then by the class equation

p ∤ [G:H| = ∑ |K| / |K ∩ H^a|
power of p since |K| = p^n

=> K ∩ H^a = K for some a $\in$ G

Since otherwise, the ∑ divisible by p whereas p does not divide [G:H]

3) cardinality
Idea: TODO

By 2), the p-Sylow subgroups are all conjugates of each other.
By the class equation with H acting on G|H,

[G:H] = |N(H) / H| + ∑ |H*aH|
= [G:H]/r + ∑ |H| / |H ∩ H^a|
power of p

=> r = 1 + r/[G:H] ∑ (power of p)
| p since p divides the sum but not [G:H]
=> r = 1 mod p

QED
==========

Theorem (various properties)
Let G be a group and p prime

|G| = p		=>	G cyclic
|G| = p^2		=>	G abelian G = Cp^2 or Cp x Cp
|G| = pq	p = 2	=>	G cyclic (C2p) or **C2 x Cq**
p < q	p ≠ 2	=>	G cyclic note: since Cp x Cq is cyclic
|G| = p^kq^l p, q are odd primes		=>	p,q-Sylow groups are normal i.e.

decomposition If all Sylow groups are normal i.e. **unique** Sylow groups Pi for each prime
then G is a direct product

G ≃ **G/P1 x G/P2 x G/P3 ...**
g <=> (gP1, gP2, gP3, ...)

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proof
- |G| = pq
If r = # q Sylow groups, then

r | p => r = 1 or p
r = 1 mod q => r = 1 if p ≠ 2 since q is always odd

Hence if p ≠ 2, then r =1

G / Cq has order p
=> G ≃ Cq x Cp ≃ Cqp since p, q are corprime

- |G| = p^2 is abelian
Consider the center ZG.
If |ZG| = p^2 then done.
Otherwise, by the class equation, |ZG| > 1 so

|ZG| = p

=> |G / ZG| = p note: ZG is normal by definition

=> G/ZG = (a) for some element a ∈ G
=> G is generated by ZG and a note: ZG commutes with a by definition

QED

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==OLD==

^: G x →

Consider the **conjugation action**

^: G x G → G
g ^ x = x^g

Then the center C(g) is precisely the **kernel of g^**

C(g) = ker g^

Restricting this to S

^S: N(S) x S → S
n ^S s = s^n note: n ∈ N(S) is required instead of any n ∈ G so that n^ is a permutation

[^S]**: N(S) → Sym(S)**
n ↦ n^: S → S

The center is precisely the **kernel** of the conjugations

C(S) := **ker [**^S**]**
= { n ∈ N(S) | n^ = 1 }
i.e. n^s = s for all s ∈ S

= { g ∈ G | S ⊆ C(G) }

n ∈ C(S) <=> n^ = 1
<=> S ⊆ C(n)

Hence C(S) is normal in N(S) and hence an isomorphism

N(S) / C(S)	≃	**Inn(S)**
i.e. N(S)/ker[^S]		i.e. Im [^S]

n → nC(S) <=> n ↦ n^

Theorem: normalizer, centralizer
Let G be a group and S ⊆ G.

a) **subgroup**: both N(S) and C(S) are subgroups of G.
S need **NOT** be a subgroup.

In particular, if g ∈ C(s), then

g^-1s = g^-1s(gg^-1)
= g^-1(gs)g^-1
= sg^-1

i.e. C(s) is closed under inversion.
Similar for C(S) and N(S)

b) equations

Then

canonical

H is normal <=> G = N(H)

H is abelian <=> H ⊆ C(H)

hierarchy

S ⊆ C^2(S) ⊆ C^4(S) ...
C(S) ⊆ C^3(S) ⊆ C^5(S) ...

C^i(S) ⊆ N^i(S) note: since C(C(T)) ⊆ N(C(T)) ⊆ N(N(T)) for any T

A special subset of N(S) is the **centralizer**

C(a) := { g ∈ G | ga = ag } i.e. commutes with a
= { g ∈ G | a^g = a }

C(S) := { g ∈ G | gs = sg for all s ∈ S } i.e. commutes with all s
= { g ∈ G | S ⊆ C(g) }
= ∩ C(s)
⊆ N(S)

The centralizer can also be thought of as the kernel of **inner maps**.

i.e. if

$\varphi n: S \to S$ where $n \in N(S)$
$\varphi n(x) = x^n$ note: NOT any $g \in G$, since it may be that $x^g \notin S$ for some x

$\varphi: N(S) \to Sym(S)$ note: Sym(S) = **symmetric** group aka bijections of S
$\varphi(n) = \varphi n$ Sym(S) = Aut(S) if S is a subgroup

Then

$C(n) \cap S = \ker \varphi n$

$C(S) = \ker \varphi$
$= \{ g \in G \mid S \subseteq C(g) \}$

$n \in C(S) \iff \varphi n = 1$
$\iff S \subseteq C(n)$

C(S) is always normal in N(S) (since kernel is always normal)

$C(S) \lhd N(S)$

==========

QED
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And

set $N^G \simeq$

class $a \to a^G$

Then conjugation is a type of **action by G** on N

conjugation action on N
*: G x N → N
g * a = a^g

a^1	=	a	identity
(a^g)^h	=	h(gag^-1)^h^-1	group operation
	=	(hg)a(hg)^-1	
	=	a^(hg)	

Then **conjugacy classes** a^G and **center C(a)** correspond to the **orbits G * a** (image of a^) and **stabilizers Ga** (kernel of a^)

a^G = **Im a^**
aka orbit G*a
= { a^g ∈ N: g ∈ G }

C(a) = **ker a^**
aka stabilizer Ga
= { g ∈ G: a^g = a }

By the property of group actions, the unique conjugates of a are characterized by the **center** (aka stabilizer, kernel of a^)

a^g = a^h <=> g ~ h mod C(a)
i.e. g = hv for some v ∈ C(a)

Hence a 1-1 relation

set a^G ≃ G|C(a)
note: C(a) need to be normal

class g ↦ a^g <=> g ↦ gC(a)

Furthermore, conjugacy classes themselves are equivalence classes

a^g = b^h <=> a ~ b mod G
i.e. y = a^g for some g ∈ G
<=> b ∈ a^G

i.e.

G = ∪ a^G i.e. disjoint union

|G| = ∑ |a^G| ∑ over unique conjugacy classes
= ∑ |G| / |C(a)|

	Action	**Class G*x**
⊆ **X**	**Permutation G*: X → X ∈ S(X)**	**Class equation**
Group on itself	*: G x G → G	image
group	g*: G → G automorphism	|G| = |G| only one orbit
	g * x = gx	G = G*x ≃ G/{1}
	g(x) = gx	gx <=> g
		# classes = 1
	image: G * x = G	class size |G*x| = |G|
	kernel: Gx = {1}	
Subgroup	*: H x G → G	image
group	h*: G → G automorphism	|G| = ∑ |gH| over unique cosets
	h * g = gh	gH ≃ H/{1}
	h(x) = hx	= ∑ |H|
	for a subgroup H	gh <=> h
		# classes = |G| / |H|
	image: H * g = gH	class size |H * g| = |H|
	kernel: Hg = {1}	

Homomorphism

*: G x H → H

f(g)*: H → H

g * H = hf(g)

f(g)(h) = hf(g)

where f: G → H is a group homomorphism

automorphism if g ∈ Ker f

image: G * h = h(Im f)

kernel: Gh = Ker f

image

|H| = ∑ |Im f| / |Ker f| over unique cosets h(Im f)

h(Im f) ≃ G/Ker f

hf(g) <=> gKer f

Conjugation

*: G x N → N

φg: N → N

g * n = n^g

φg(x) = x^g = gng^-1

for a normal subgroup N

image

|N| = ∑ |n^G| over unique n^G

n^G ≃ G|C(n)

n^g <=> gC(n)

classes = ?

class size |G*n| = |G| / |C(n)|

Image: G*n = n^G

Kernel: Gn = C(n)

Left cosets G|H

*: G x G|H → G|H

g: G|H → GH

g * aH = gaH

g(aH) = gaH

image

|G|H| = |G|/|H|

G|H ≃ G|H^a

g(aH) <=> g(H^a) = 1

classes

Image: G * aH = G|H

kernel: GaH = H^a

class size |G * aH| = |G| / |H|

Conjugacy classes H^G

*: G x H^G → H^G

g * H^a = H^(ga)

image

|H^G| = |G|N(H)|

G|N(H) ≃ K^g <=> gN(K)

= | G| / |N(H)|

where K = H^a

G|N(K)

Image: G * H^a = H^G

Kernel: GH^a = N(H^a)

classes = 1

class size |G *

H^a| = |G| / |N(H)|

H^G ≃ G|N(H)
H^g <=> gN(H)

Theorem: Normal Lattice
Let H, N be subgroups of G

a) **well-defined**:
HN is a subgroup of G
HN = { hn: h ∈ H, n ∈ N } is a subgroup of G if N is normal

b) **normality**: if H and N are normal, then H∩N and HN are normal

c) **lattice**: normal subgroups form a **lattice** by subset inclusion ⊆
i.e. for any normal subgroup N, H

==========
Proof:
a)
- HN is a subgroup:
Suppose N is normal. Then for any hn, h'n' ∈ HN

1) product: for any hn, h'n' ∈ HN

(hn)(h'n') = h(h'n'')n' nh' = h'n'' for some n'' by normality
= (hh')(n''n') ∈ HN

2) inverse: let hn ∈ HN, so its inverse is n^-1 h^-1
Then

n^-1 h^-1 = h^-1n' for some h' since h^-1N = Nh^-1 by normality
∈ HN

b)
- HN is normal: let hn ∈ HN and g ∈ G

=> HN^G = HN

QED
==========

All have the same order

|H| = |aH| = |H^a|

The cosets and conjugates are closely related

abelian H^a = H <=> aH = Ha

Theorem: left cosets
Let H be a subgroup of G

a) **element**: if b ∈ aH, then

order |H| = |aH|

[G:H] = |G|H|

= |G| / |H|

representative bH = aH <=> b ∈ aH

i.e. any b ∈ aH can be used as a **representative** of aH.

note: =>) by if b ∈ aH, then b = ah for some h

identity b(aH) = aH <=> b ∈ H^a

1-1 correspondence

G * aH ≃ G|H^a

b(aH) <=> b(H^a)

b(aH) = H <=> b ∈ H(a^-1)

If N is a **normal** subgroup, then this is just

b(aN) = aN <=> b ∈ N

since N^a = N

b(aN) = N <=> b ∈ a^-1N

since gN = Ng

weak commutative aH = Ha <=> H^a = H

i.e. **invariance** under conjugation is equivalent to weak commutative

power (a^k)H = (a^l)H <=> a^l-k ∈ H

Hence, if a^n ∈ H is the smallest such n, then

` aH, a^2H, ..., a^n-1 H, a^nH = H

are **unique**.

c) **order**: cosets have the same **order**

In particular

|aH| = |H| since each ah is unique for each h ∈ H

Thus, if G is finite, the number of cosets is

[G:H] = |G| / |H|

[G:H] is the **index of H** in G

d) **equivalence**: cosets are equivalence classes mod H

i.e.

x ∈ aH <=> a ~ x mod H

x^-1a ∈ H

e) cosets are **disjoint**
since they are equivalence classes



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